

$$\mathcal{L}_e = \mathcal{L}_{\text{MSE}} + \lambda_s \mathcal{L}_{\text{slope}} + \lambda_p(e) \mathcal{L}_{\text{peak}} + w(e) \mathcal{L}_{\text{inv}} + 0.1 \mathcal{L}_{\text{lat}}, \quad \lambda_s = 0.5$$

$$\lambda_p(e) = 0.05 \text{ if epoch } e > 0.8 \times \text{total epochs, else } 0, \quad w(e) = \min\left(1, \frac{e+1}{10}\right)$$

$$\mathcal{L}_{\text{MSE}}, \mathcal{L}_{\text{slope}}, \mathcal{L}_{\text{peak}} \text{ on the ERP } \hat{y} = \Psi \mathbf{b}(\mathbf{a}) + \psi_0, \text{ as in the forward loss}$$

$$\mathcal{L}_{\text{inv}} = \text{Huber}(\hat{\mathbf{a}}, \mathbf{a}), \quad (\hat{\mathbf{a}}, \hat{\mathbf{z}}) = T^{-1}(\mathbf{b}^*), \quad \mathbf{b}^* = \frac{1}{F} \Psi^\top (y - \psi_0)$$

$$\mathcal{L}_{\text{lat}} = \text{Huber}(\hat{\mathbf{z}}, \mathbf{0})$$